

Amorphous Cell Construction in PAAF

Theory, Algorithm, Implementation and Validation

Polymer Auto-Assembly Framework (PAAF)

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Abstract

PAAF builds periodic amorphous polymer cells by *chain growth*: each chain is constructed one skeletal bond at a time, with the torsion about that bond drawn from a rotational isomeric state (RIS) distribution reweighted by the non-bonded environment already present in the box. This is the method of Theodorou and Suter [1], optionally combined with the scanning method of Meirovitch [2], on the statistical foundation laid by Flory [3]. It is the same published pair of algorithms that underlies the BIOVIA Materials Studio *Amorphous Cell* module, which BIOVIA documents as “the combined use of the arc algorithm developed by Theodorou and Suter and the scanning method of Meirovitch” [10].

This report states the theory with its equations, specifies exactly what PAAF implements, reports the measured validation of every physical claim, and sets out — without softening — what the implementation does *not* do. The characteristic ratio C_∞ is reproduced within 12% of tabulated values for all four parameterised polymers; bond lengths and angles are exact to floating-point precision; melt densities are reached, unbiased at 0.85 and 1.04 g/cm³. The principal limitations are that PAAF performs *construction only* with no subsequent energy minimisation, that growth is at united-skeletal-atom resolution followed by back-mapping rather than direct full-atomistic construction, and that polymer-specific RIS parameters exist for four polymers only. Section 9 assesses novelty honestly: the method is a faithful reimplementa-
tion, not a new algorithm.

Contents

1	Why growth rather than insertion	2
2	Rotational isomeric state theory	2
2.1	The three-state model	2
2.2	Exact finite-chain sampling	3
2.3	Chain geometry	3
2.4	The characteristic ratio, and why it is the test	4
3	Growth into the periodic cell	4
3.1	The growth probability	4
3.2	The non-bonded term	4
3.3	Hard rejection: overlap and spearing	5
3.4	Meirovitch scanning and the Rosenbluth bias	6
3.5	Growth granularity: one bead per skeletal atom	6
3.6	Repeat-unit mass from SMILES	7
3.7	Growth order	7
4	Composition: from formulation to cell	8
5	From skeletal beads to atoms	8
6	The algorithm, in full	10

7	Validation	10
7.1	Chain dimensions against literature	10
7.2	Convergence and internal consistency	10
7.3	Coverage across the polymer library	11
7.4	Cell-level checks	11
8	Limitations	11
9	Relation to existing tools, and an honest novelty assessment	13
9.1	Comparison	13
9.2	Novelty	13

1 Why growth rather than insertion

The natural first approach to filling a periodic box with polymer is to build whole chains and insert them at random positions and orientations, rejecting overlaps. This works for solvents and short oligomers and fails for polymers, for a reason that is geometric rather than a matter of tuning.

A random coil pervades a volume set by its radius of gyration, $R_g \propto N^{1/2}$, so the pervaded volume scales as $N^{3/2}$ while the mass inside it scales only as N . At melt density that pervaded volume is mostly *other chains*: a polymer coil is roughly 99% not-itself. Inserting a chain whole therefore requires locating a cavity shaped like an entire coil, and the probability of that decreases roughly exponentially in N . Acceptance collapses somewhere between 20 and 50 repeat units.

Growing the chain bond by bond replaces one impossible question — “is there a coil-shaped hole?” — with many easy ones — “is there room for one more skeletal atom?”. Each step requires only local free volume, and the chain finds its own way through the material already placed. This is the central insight of Theodorou and Suter [1], and it is why every serious amorphous cell builder grows rather than inserts.

2 Rotational isomeric state theory

2.1 The three-state model

A backbone torsion is not free. Rotation about a C–C bond has three minima — *trans* ($\phi \approx 180^\circ$) and *gauche* $^\pm$ ($\phi \approx \pm 60^\circ$) — separated by barriers of a few kcal/mol. The RIS approximation of Flory and Volkenstein replaces the continuous torsional potential by these discrete states [3, 4]. Two facts follow, and both control chain dimensions.

First, the minima are not equally weighted. A *gauche* state costs an extra energy E_σ relative to *trans*, giving the first-order statistical weight

$$\sigma = \exp\left(-\frac{E_\sigma}{RT}\right) < 1. \quad (1)$$

Second, neighbouring torsions are *correlated*. A g^+g^- pair drives the two chain segments on either side into one another — the *pentane effect* — and carries a large additional penalty E_ω ,

$$\omega = \exp\left(-\frac{E_\omega}{RT}\right) \ll 1. \quad (2)$$

Both are encoded in the 3×3 statistical weight matrix, indexed $U[\text{previous state}, \text{current state}]$ in the order (t, g^+, g^-) :

$$\mathbf{U}(T) = \begin{pmatrix} 1 & \sigma & \sigma \\ 1 & \sigma & \sigma\omega \\ 1 & \sigma\omega & \sigma \end{pmatrix}. \quad (3)$$

Polymer	State	Realised	Predicted	Δ
PE	t	0.6011	0.5974	+0.0037
	g^+	0.2036	0.2013	+0.0023
	g^-	0.1953	0.2013	-0.0060
PS	t	0.6786	0.6741	+0.0045
	g^+	0.1629	0.1630	-0.0001
	g^-	0.1585	0.1630	-0.0044

Table 1: Sampled torsion-state populations against the transfer-matrix prediction, $T = 413$ K, 80 chains of 500 torsions. Measured by `tests/test_ris.py`.

Every row begins with 1 because trans is the reference state. The factor ω appears only on the g^+g^- and g^-g^+ entries — exactly the pairs whose substituents collide. In PAAF at $T = 413$ K for polyethylene this gives $U[g^+, g^+] = 0.5438$ against $U[g^+, g^-] = 0.0421$, a suppression of $12.9\times$, which is the pentane effect made numerical.

2.2 Exact finite-chain sampling

A naive sampler would draw the next state from $P(k) \propto U[j, k]$. That is wrong for a chain of finite length, because it ignores how many conformations each choice leaves available downstream, and it biases the chain ends.

PAAF instead uses Flory’s matrix method as a *sampler*. The partition function is accumulated backwards from the far end of the chain,

$$\mathbf{Z}^{(i)} = \mathbf{U} \mathbf{Z}^{(i+1)}, \quad \mathbf{Z}^{(n)} = \mathbf{1}, \quad (4)$$

after which the forward conditional probability is exact:

$$P(k | j) = \frac{U[j, k] Z_k^{(i+1)}}{\sum_{k'} U[j, k'] Z_{k'}^{(i+1)}}. \quad (5)$$

The recursion in Eq. (4) is renormalised at each step, since the product underflows for long chains and only ratios are ever used.

Table 1 shows that the realised state populations from Eq. (5) reproduce the transfer-matrix prediction — the exact long-chain limit obtained from the principal eigenvector of $\mathbf{U}$ — to better than 0.006 in every state.

2.3 Chain geometry

Bond lengths l and bond angles θ are held *fixed*; only the torsion varies. Each successive skeletal atom is placed by exact internal coordinates using the Natural Extension Reference Frame construction: given three consecutive placed atoms $\mathbf{a}, \mathbf{b}, \mathbf{c}$, the next atom is

$$\mathbf{d} = \mathbf{c} + l \mathbf{R}(\mathbf{a}, \mathbf{b}, \mathbf{c}) \begin{pmatrix} -\cos \theta \\ \sin \theta \cos \phi \\ \sin \theta \sin \phi \end{pmatrix}, \quad (6)$$

where $\mathbf{R}$ is the orthonormal frame built from the preceding three atoms. Because Eq. (6) is an exact construction rather than a minimisation, the geometry cannot drift: PAAF measures bond lengths of $1.530000 \pm 1.5 \times 10^{-15}$ Å and bond angles of $112.0000 \pm 5.7 \times 10^{-14}^\circ$ over chains of 300 torsions, which is floating-point noise.

2.4 The characteristic ratio, and why it is the test

The observable that reveals whether an RIS implementation is real is the characteristic ratio

$$C_n = \frac{\langle R^2 \rangle}{n l^2}, \quad (7)$$

the mean-square end-to-end distance relative to a freely jointed walk of the same bond count. Its limiting value C_∞ is tabulated experimentally for real polymers. Three regimes must be distinguishable:

Model	C for PE	What it captures
Freely jointed	1.0	nothing
Freely rotating (θ fixed)	≈ 2.2	bond angle only
RIS (θ fixed, ϕ weighted)	≈ 6.7	+ torsional statistics

The freely rotating limit has a closed form that serves as an independent check on the geometry code alone:

$$C_\infty^{\text{fr}} = \frac{1 + \cos \alpha}{1 - \cos \alpha}, \quad \alpha = 180^\circ - \theta. \quad (8)$$

For $\theta = 112^\circ$ this gives 2.198; PAAF measures 2.445 ± 0.098 with torsions sampled uniformly, confirming that the chain builder is correct before any statistics are applied. With RIS weights switched on, the same code gives 6.82 against 2.22 for random torsions — a factor of 3.07. An implementation in which RIS were decorative would sit at the freely rotating baseline.

3 Growth into the periodic cell

3.1 The growth probability

The Theodorou–Suter construction adds one skeletal bond at a time. For each candidate rotational isomeric state s the trial atom position is computed from Eq. (6), and the state is chosen with probability

$$P(\phi_s) \propto U[j, s] \exp\left(-\frac{E_s^{\text{nb}}}{k_B T}\right), \quad (9)$$

where j is the state of the previous torsion and E_s^{nb} is the non-bonded energy of the trial atom against the atoms of *previously completed chains*, evaluated under the minimum-image convention within the neighbour-list cutoff. Equation (9) is the heart of the method: the first factor makes the coil the right *size*, the second makes it avoid *other chains*. Note that the growing chain’s own beads do not enter E^{nb} — a chain is inserted into the neighbour grid only once complete — so intra-chain crowding is handled entirely by the hard self-avoidance test described below.

3.2 The non-bonded term

A full Lennard-Jones sum would be the faithful choice, but it requires per-atom LJ parameters that are not available until the force field is assigned — which in PAAF happens *after* the cell is built. PAAF therefore uses a purely repulsive soft-core term keyed on van der Waals contact distance,

$$E^{\text{nb}} = \epsilon \sum_j \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12}, \quad r_{ij} < \sigma_{ij}, \quad (10)$$

with $\epsilon = 0.2$ kcal/mol.

Deviation from the paper, stated plainly. Equation (10) reproduces the qualitative effect that matters for construction — candidates that bury the new atom in existing material are strongly disfavoured — without claiming a quantitative energy. Absolute values of E^{nb} are not physical energies and are not reported as such anywhere in PAAF.

3.3 Hard rejection: overlap and spearing

Two hard tests are applied before Eq. (9) is evaluated, because no statistical weight however favourable should be able to produce them.

Overlap. A candidate is rejected if it comes closer to any non-bonded atom than

$$r_{\min} = \max\left(d_{\text{tol}}, \frac{1}{2}(R_i + R_j)s\right), \quad (11)$$

with interpenetration factor $s = 0.55$ and hard tolerance $d_{\text{tol}} = 1.7 \text{ \AA}$ by default. Neighbour searching against other scan chains uses a cell list, so that part of the cost is $O(N)$ rather than $O(N^2)$; the self-avoidance scan along the growing chain is not, and is $O(n)$ per candidate. The 1–2, 1–3 *and* 1–4 partners along the chain are excluded — the first two are at bonded distance by construction, and the 1–4 separation is set by the very torsion being chosen, so a gauche state legitimately brings it inside the tolerance. Against its own chain the test uses d_{tol} alone; the radius term in Eq. (11) applies to inter-chain contacts, and only exceeds $1.7 \text{ \AA}$ above roughly 226 amu per bead, which none of the polymers in Table 3 reach.

Ring spearing. A threaded ring is a topological defect: the bond cannot be pulled back out, so energy minimisation relaxes the strain and leaves the two permanently interlocked. It is a classic silent failure of constructed cells. Each ring is approximated by the disc spanned by its atoms and the segment–disc intersection solved directly,

$$t = \frac{\mathbf{n} \cdot (\mathbf{c}_0 - \mathbf{p}_0)}{\mathbf{n} \cdot (\mathbf{p}_1 - \mathbf{p}_0)}, \quad \text{speared} \iff 0 \leq t \leq 1 \wedge \|\mathbf{h} - \mathbf{c}_0\| \leq R_{\text{ring}}, \quad (12)$$

where $\mathbf{h}$ is the crossing point.

Where this test can and cannot run. At skeletal-bead resolution a pendant ring is not represented at all — polystyrene’s phenyl group does not exist until back-mapping places it — so during growth there is literally nothing to spear and the predicate, though implemented and unit-tested against four geometric cases, never fires. PAAF therefore performs the check on the finished *all-atom* cell instead (Section 5), where it is a diagnostic rather than a rejection criterion. This is worth doing: applying it to a polystyrene cell at 1.04 g/cm^3 immediately found **3 speared bonds** among 20 phenyl rings — a defect the construction was previously blind to. A full-atomistic grower could reject these during growth; PAAF can only report them.

On the overlap tolerance. d_{tol} is *not* the physical contact distance and must not be set to one. Equation (10) already discourages close approach; d_{tol} is only a guard against an overlap so severe that no subsequent minimisation could repair it. Setting it to the true united-atom contact distance double-counts the repulsion and prevents the cell being built at all. The default was measured, not chosen: see Table 2.

d_{tol} (Å)	Cells built	Mean C_n
2.0	1 / 3	5.68
1.9	2 / 3	6.69
1.8	2 / 3	7.60
1.7	3 / 3	5.74
1.5	3 / 3	5.27

Table 2: Effect of the hard overlap tolerance. Eight polyethylene chains of DP = 60 grown into a 29.7 Å box at 0.85 g/cm³ (the melt density at 413 K), unbiased sampling, three random seeds each. Increasing `max_restarts` instead is a worse remedy: at 2.0 Å, raising restarts from 30 to 150 still reached only 2/3 and pulled C_n down to 4.97, because surviving a restart selects for compact chains.

3.4 Meirovitch scanning and the Rosenbluth bias

Growth can walk into a dead end from which no continuation is acceptable, and the whole chain is then lost — *attrition*. The scanning method of Meirovitch [2] looks k steps ahead and weights each candidate by the total surviving weight of its continuations,

$$W_s^{(k)} = \sum_{\text{continuations } c} \prod_{m=1}^k U[s_{m-1}, s_m] \exp\left(-\frac{E_m^{\text{nb}}}{k_B T}\right), \quad (13)$$

so that Eq. (9) is replaced by $P(\phi_s) \propto U[j, s] \exp(-E_s^{\text{nb}}/k_B T) W_s^{(k)}$. The cost is 3^k per step, so $k \in \{1, 2, 3\}$ is the practical range.

Scanning demonstrably works against attrition. Four polyethylene chains of DP = 40 at 1.10 g/cm³ over three seeds: $k = 0$ built 2 of 3 cells with 23 restarts; $k = 2$ built 3 of 3 with 3 restarts.

Scanning is off by default, and the reason is measured. Selecting a state in proportion to its look-ahead weight is a *biased* estimator — the Rosenbluth bias — because states that leave more room are over-chosen, and those are the extended ones. Measured on dilute polyethylene (2 chains of DP = 40 at 0.10 g/cm³, 12 seeds), against $C_n = 6.78 \pm 0.24$ for the same chain grown unperturbed: 5.66 ± 0.74 at $k = 0$, 7.40 ± 1.07 at $k = 1$, 7.52 ± 1.35 at $k = 2$. Scanning overshoots the unperturbed dimensions; $k = 0$ undershoots them. Removing the bias requires carrying the Rosenbluth weight

$$W = \prod_i \sum_s U[j_i, s] \exp(-E_{i,s}^{\text{nb}}/k_B T)$$

through the construction and using it in every subsequent average, which is *not* implemented in PAAF. Therefore: use $k = 0$ when chain dimensions matter; use $k > 0$ only when attrition would otherwise prevent the cell being built at all, and do not quote C_n from such a cell. PAAF attaches this warning to any result produced with $k > 0$.

3.5 Growth granularity: one bead per skeletal atom

The RIS model parameterises a *skeletal bond*. A growth step must therefore be a skeletal bond, not a repeat unit. PAAF determines the number of skeletal atoms a repeat unit contributes by finding the shortest path between the two polymerisation attachment points in the SMILES graph:

$$n_{\text{skel}} = |\text{path}(a_1 \rightarrow a_2)|, \quad a_k = \text{heavy atom bonded to } *_k, \quad m_{\text{bead}} = \frac{M_{\text{unit}}}{n_{\text{skel}}}. \quad (14)$$

Side groups are branches off that path and do not count. Table 3 gives the resulting values.

Polymer	Repeat unit SMILES	n_{skel}	M_{unit}	m_{bead}
Polyethylene	<chem>[*]CC[*]</chem>	2	28.05	14.03
Polypropylene	<chem>[*]CC([*])C</chem>	2	42.08	21.04
Polystyrene	<chem>[*]CC([*])c1ccccc1</chem>	2	104.15	52.08
Poly(vinyl chloride)	<chem>[*]CC([*])Cl</chem>	2	62.50	31.25
Poly(ethylene oxide)	<chem>[*]OCC[*]</chem>	3	44.05	14.68
Poly(butylene succinate)	<chem>[*]OCCCCOC(=O)CCC(=O)[*]</chem>	10	172.18	17.22

Table 3: Skeletal granularity and masses derived from SMILES, in g/mol. Note that the phenyl ring of polystyrene is a branch and correctly contributes 0 to n_{skel} .

This was a real defect, and it is worth recording why. An earlier version of PAAF placed one bead per *repeat unit* at the RIS bond length 1.53 Å. Density came out correct, because density knows only about mass and volume. But the contour length was too short by a factor of n_{skel} , so every dimension derived from the chain path — R_g , end-to-end distance, C_n — was too small by $\sqrt{n_{\text{skel}}}$: $\sqrt{2}$ for the vinyl polymers, $\sqrt{10}$ for poly(butylene succinate). A correct-looking density concealed it. The lesson generalises: check an observable that depends on the chain *path*, not only one that depends on its mass.

3.6 Repeat-unit mass from SMILES

All masses are computed from the SMILES with implicit hydrogens made explicit, and with one correction that is easy to get wrong. Stripping the [*] markers leaves a *closed-shell* molecule: [*]CC[*] becomes CC, which a cheminformatics toolkit dutifully caps with hydrogen, giving ethane C2H6. But in a polymer those two carbons bond to neighbouring repeat units, not to hydrogen. The repeat unit is C2H4. PAAF therefore subtracts one hydrogen per attachment point,

$$M_{\text{unit}} = \sum_e n_e m_e - n_* m_{\text{H}}. \quad (15)$$

Without this correction every mass is high by 1.008 per connection — 2.016 g/mol for a linear unit, a 7% error for polyethylene — which would propagate directly into cell density and into every weight fraction.

3.7 Growth order

Chains enter the box one at a time, so whichever species is grown *last* faces a box already at its final density. Taking species in the order the user happens to list them therefore makes success depend on an arbitrary choice: a 70/30 PE/PS blend fails on the very first polystyrene chain, because 27 polyethylene chains have already filled the box and a polystyrene bead has radius 2.74 Å against polyethylene’s 1.77.

PAAF therefore chooses the order by difficulty,

$$\text{rank} = n_{\text{beads}} \times R_{\text{bead}}^3, \quad \text{descending}, \quad (16)$$

so the hardest species claims the empty box and the easiest fills the gaps. The chosen order is reported in the result whenever it differs from the one the caller supplied. Atoms are *assembled* in the caller’s original order regardless, so that downstream consumers which slice the atom list by species are unaffected.

This reordering is an engineering addition by PAAF, not part of Theodorou–Suter or Meirovitch, and is identified as such.

Target beads	Chains	Realised wt% PE	Error (pp)
400	5	51.86	18.14
2 000	20	70.80	0.80
10 000	100	70.80	0.80
40 000	400	70.22	0.22

Table 4: Composition error against cell size for a requested 70/30 PE/PS blend at DP = 50. The error is a property of integers, not a defect; PAAF states it rather than concealing it.

4 Composition: from formulation to cell

A formulation is specified in weight percent; the grower needs integer chain counts and a box edge. PAAF solves this explicitly rather than implicitly.

Given target weight fractions w_i and chain masses $m_i = M_{\text{unit},i} \text{DP}_i$, the ideal fractional chain count is

$$n_i = \frac{w_i M_{\text{total}}}{m_i}, \quad (17)$$

and the integers are chosen by largest remainder: floor everything, then allocate the leftover chains to whichever species is furthest below its target. Every species requested with non-zero weight receives at least one chain — a “5 wt% additive” that rounds to zero chains is not a 5 wt% cell, and silently dropping it would be the worst available outcome.

Dividing by m_i in Eq. (17) is essential and is the step most often got wrong. A DP = 50 polystyrene chain is 5 208 g/mol against 1 403 for polyethylene. Allocating chains in proportion to weight percent directly would give 7 : 3 for a 70/30 blend and a realised composition of 38.6 wt% PE. The correct solution is 18 : 2, giving 70.80 wt%.

The box edge follows from the target density ρ :

$$a = \left(\frac{M_{\text{total}}}{N_A \rho} \right)^{1/3} \times 10^8 \text{ \AA}. \quad (18)$$

Because chain counts are integers, the realised composition is generally not the requested one. PAAF reports both, side by side, and states the error in percentage points rather than rounding silently. The error falls with cell size as Table 4 shows.

5 From skeletal beads to atoms

Growth produces skeletal atoms. No force-field typer can work with those: DL-FIELD and Moltemplate need elements, hydrogens, side groups and bond orders. PAAF therefore back-maps, and it can do so cleanly *because* a bead is a skeletal atom — the backbone of the atomistic chain already exists, exactly, at positions the growth accepted.

For each repeat unit PAAF embeds one 3D template: the repeat unit with its two attachment points converted to hydrogens, minimised with MMFF (falling back to UFF where MMFF typing fails). In that template each backbone atom has a local frame defined by its two backbone neighbours (a link hydrogen standing in at the unit ends). The same frame is built at the corresponding bead in the grown chain, and the rigid transform between them,

$$\mathbf{R} = \mathbf{F}_{\text{target}} \mathbf{F}_{\text{template}}^T, \quad \mathbf{x}'_k = \mathbf{p}_{\text{bead}} + \mathbf{R}(\mathbf{x}_k - \mathbf{x}_{\text{backbone}}), \quad (19)$$

is applied to everything attached to that backbone atom. Because Eq. (19) is rigid, every substituent bond length and bond angle survives exactly.

Verification: back-mapped formulae are exact — a DP = n polyethylene chain gives $\text{C}_{2n}\text{H}_{4n+2}$ and a polystyrene chain $\text{C}_{8n}\text{H}_{8n+2}$, confirmed for multi-chain cells — and the backbone atoms land on the grown beads to within $1.8 \times 10^{-15} \text{ \AA}$.

Push-off. A hydrogen sits 1.09 Å from its carbon, so two backbone atoms at an acceptable separation can still leave their hydrogens nearly coincident. Handing a 0.3 Å contact to LAMMPS is how a minimisation explodes. PAAF therefore applies a purely geometric relief step in which *only non-backbone atoms move*. It has two parts, and the order matters.

First, each side group is **rotated rigidly** about the single bond joining it to the backbone. That rotation is exact — every bond length, bond angle and internal torsion of the group is preserved, because the group moves as a body — and it is the *only* move that can help a ring. It is precisely the side-group torsion that growth does not sample, recovered here at negligible cost. Each rotatable group is scanned over 36 uniform angles and placed where a soft repulsive score against everything outside the group is smallest.

Second, the remaining atoms are pushed apart, after which a SHAKE-style constraint relaxation restores **every** bond, including ring closures, to its original length.

Measured: polyethylene 0.34 → 1.65 Å (no rotatable groups); polystyrene 0.31 → 1.00 Å with 20 phenyl groups rotated. Bond lengths after both steps span 1.087–1.530 Å for polystyrene and 1.094–1.530 Å for polyethylene, i.e. unchanged. Bond angles and inter-group torsions do drift; that is the price, and it is the cheap part of the geometry for a real minimiser to repair.

Bug history, recorded because the failure mode is instructive. An earlier implementation pinned each mobile atom to a *single* parent and re-projected onto that one bond. A spanning tree cannot constrain a cycle, so the ring-closing bond of every phenyl group was left completely free and was torn open to as much as 5.0 Å — producing a structure that looked plausible and was chemically destroyed. Nothing caught it, because the only bond-length test in the suite used polyethylene, where every mobile atom is a hydrogen bonded directly to a pinned backbone carbon and a tree is therefore sufficient. *Test the case that contains the cycle.*

Spearing, checked here. Once the rings exist, the spearing test of Eq. (12) can finally be applied. Rings are located by a depth-limited search for the shortest alternative path across each bond, then approximated as discs by singular-value decomposition of their atom positions. Every bond in the cell is tested against every ring under minimum image. A polystyrene cell at 1.04 g/cm³ contained 20 phenyl rings and **3 speared bonds**; a polyethylene cell contains no rings and reports none. The count is attached to the result and raised as a warning.

Tacticity. Applying one template to every unit produces an *isotactic* chain. This is a real stereochemical choice that changes properties, so PAAF exposes it rather than hiding it: **syndiotactic** alternates and **atactic** mirrors each unit's substituents through the local backbone plane at random, inverting the configuration at the stereocentre.

Polymer	E_σ	E_ω	θ	C_∞ meas.	Ref.	Dev.	Source of parameters
PE	0.50	2.10	112°	6.26 ± 0.25	6.7	−6.6%	Flory 1969 [3]
PP	0.20	1.65	114°	5.83 ± 0.22	5.5	+5.9%	Suter & Flory 1975 [5]
PS	0.85	2.40	114°	8.90 ± 0.38	10.0	−11.0%	Yoon <i>et al.</i> 1975 [6]
PMMA	0.70	2.20	116°	9.16 ± 0.38	8.2	+11.8%	Sundararajan & Flory 1974 [7]

Table 5: Characteristic ratio against tabulated values. Energies in kcal/mol, $T = 413$ K, 400 chains of 500 bonds each. The PP, PS and PMMA entries average over tacticity rather than modelling the meso/racemo distinction explicitly, which is the principal source of their deviation.

6 The algorithm, in full

```

1: input: repeat-unit SMILES, DP and wt% per species; target density  $\rho$ ; temperature  $T$ 
2: compute  $M_{\text{unit}}$  (Eq. 15) and  $n_{\text{skel}}$  (Eq. 14) for each species
3: solve integer chain counts (Eq. 17) and cubic box edge  $a$  (Eq. 18)
4: order species by difficulty (Eq. 16)
5: for each species  $i$  in difficulty order:
6:   for each chain of species  $i$ :
7:     for attempt = 1 ...  $N_{\text{restart}}$ :
8:       seed three atoms at a random position having local free volume
9:       while chain incomplete:
10:        for each rotational isomeric state  $s \in \{t, g^+, g^-\}$ :
11:          place trial atom by NeRF (Eq. 6)
12:          reject if overlap (Eq. 11) or ring spearing (Eq. 12)
13:          evaluate  $E_s^{\text{nb}}$  (Eq. 10); optionally look ahead (Eq. 13)
14:           $w_s \leftarrow U[j, s] \exp(-E_s^{\text{nb}}/k_B T)$ 
15:          if all  $w_s = 0$ : restart chain (dead end — attrition)
16:          else: sample  $s$  from  $w_s / \sum w$ ; accept the atom
17:          wrap the chain, insert into the neighbour cell list, record  $R_g$ ,  $R_{ee}$ ,  $C_n$ 
18:          assemble atoms in the caller's species order, not the growth order
19:          back-map to all-atom (Eq. 19); push off side-group overlaps
20:          assign atom types; export LAMMPS via DL_FIELD or Moltemplate

```

Algorithm 1: PAAF amorphous cell construction.

7 Validation

Every physical claim PAAF makes is checked against an independent reference, and the measured value is printed rather than merely asserted. The suite comprises 442 tests, most of them a sweep of the whole polymer library; the GUI subset additionally requires PyQt5.

7.1 Chain dimensions against literature

Table 5 is the load-bearing result. Deviations fall in both directions, which is what one expects from a genuine model rather than a fitted fudge factor.

7.2 Convergence and internal consistency

C_n must increase with chain length and level off, not sit flat. Measured for PE at 413 K: 5.01 ($n = 20$), 6.33 (50), 6.30 (100), 7.04 (200), 7.21 (400). Cooling must favour trans, and does: the trans fraction rises $0.573 \rightarrow 0.602 \rightarrow 0.643$ as T falls $500 \rightarrow 400 \rightarrow 300$ K.

7.3 Coverage across the polymer library

Validating two polymers demonstrates that two polymers work. Every entry in PAAF’s 124-polymer library is therefore driven through chemistry, growth and back-mapping, and each must either yield sound bond lengths or be refused with a specific reason. The sweep found real defects that PE and PS could not have exposed:

- **Poly(methyl methacrylate) could not be parsed at all.** Its repeat unit [*]CC([*])(C)C(=O)OC has a quaternary backbone carbon, so the tail after the second marker *already* begins with a branch. Wrapping it in another pair of parentheses produced CC((C)C(=O)OC) — a doubled bracket that is not valid SMILES. One of the four polymers whose RIS parameters this report tabulates could not have its mass computed. Fixed by peeling the leading balanced branches before parenthesising the remainder; PMMA now gives 100.12 g/mol ($C_5H_8O_2$), exact.
- **Poly(dimethylsiloxane) crashed.** Both attachment points sit on the same silicon, and RDKit raises an invariant violation when asked for the shortest path from an atom to itself. The single-skeletal-atom case is now handled directly; PDMS gives $Si_{16}C_{32}H_{100}$ for 2 chains of $DP = 8$, exact.
- **24 polymers have a ring in the backbone** and were silently producing torn structures with bonds up to 11.9 Å. They are now refused with a specific explanation (limitation L9).
- **One library entry, MAH, carries a non-polymerisable SMILES** in its polymerisation field. This is a data defect rather than a code defect and is asserted against so it cannot spread unnoticed.

7.4 Cell-level checks

- **Density.** Target densities are reached exactly: 0.8500, 0.8800, 0.9000, 0.9500, 1.0400 g/cm³ all realised to four decimal places, because the box is sized analytically from Eq. (18). Of these, 0.85 and 1.04 are reached with unbiased sampling; 0.88, 0.90 and 0.95 come from tests that enable $k = 1$.
- **Geometry preserved under growth.** Bond length $1.530000 \pm 1.4 \times 10^{-15}$ Å, angle $112.0000 \pm 5.1 \times 10^{-14}^\circ$ in a packed cell.
- **No overlaps.** Minimum non-bonded distance under minimum image measured at 2.002 Å for a requested tolerance of 2.0 Å.
- **Spearing predicate.** Unit-tested against four geometric cases: straight through, passing beside, parallel to the ring plane, and stopping short.
- **Reproducibility.** Identical seed gives identical coordinates to 0.00×10^0 Å maximum difference.
- **Melt-level C_n .** Polystyrene in a cell at 0.95 g/cm³ gives 11.13 against a reference C_∞ of 10.0.

8 Limitations

This section exists because a reader will ask these questions, and answering them first is stronger than being caught by them.

- L1. Construction only — there is no energy minimisation.** BIOVIA describes Amorphous Cell as two steps: an initial guess structure, *followed by relaxation to a state of minimum potential energy* [10]. PAAF performs only the first. The geometric push-off of Section 5 makes the structure safe to hand to a minimiser; it is not itself a minimisation and involves no force field. *Users must minimise and equilibrate in LAMMPS before quoting any*

property from a PAAF cell. This is the single largest gap between PAAF and a production tool.

- L2. Growth is at united-skeletal-atom resolution.** Theodorou and Suter place every pendant atom and sample side-group torsions for vinyl polymers. PAAF grows the backbone and reconstructs side groups afterwards from a single template rotamer, then rotates each group about its own bond to relieve overlap. That rotation recovers one degree of freedom exactly, but it is a greedy clash-relief step, *not* Boltzmann sampling of the side-group torsion: the resulting rotamer distribution has no claim to be the equilibrium one. Properties that depend on side-group packing should not be taken from an unequilibrated PAAF cell.
- L3. Spearing is detected but cannot be prevented.** Because rings do not exist until back-mapping, a speared ring can only be reported after the fact, not rejected during growth. This is not hypothetical: a polystyrene cell at 1.04 g/cm³ contained 3 speared bonds among 20 rings. PAAF raises a warning; the user must rebuild with a different seed or at lower density. A full-atomistic grower would reject these as they arise.
- L4. The non-bonded term is generic, not the production force field.** Equation (10) is a soft repulsion, not the Lennard-Jones and electrostatic terms of the force field that will later be applied. Materials Studio uses the real force field during growth.
- L5. RIS parameters exist for four polymers.** PE, PP, PS and PMMA carry literature parameterisations. Every other polymer falls back to a generic three-state model, which PAAF labels `is_generic` and flags in the result as “chain dimensions are qualitative, not a quantitative prediction”. It is honest, but it means the RIS advantage does not extend to arbitrary chemistry. This is the most consequential limitation for a user working outside the four.
- L6. Tacticity is averaged in the RIS parameters.** PP, PS and PMMA use a single averaged state set in place of the meso/racemo distinction of the published treatments, which is the main reason PMMA reads +11.8%.
- L7. Scanning is biased and therefore off by default.** See Section 3. The Rosenbluth weight is not carried, so $k > 0$ cells must not be used to quote chain dimensions.
- L8. Sequential construction over-extends chains by roughly 20%.** Chains grow into a box that starts empty, so later chains encounter free volume more open and more connected than a real melt’s, and Eq. (9) then favours the extended conformations that fit it. Measured for PE at 0.85 g/cm³, DP = 60: $C_n = 8.79$ in the cell against 7.26 for the same chain grown in isolation. Real melts screen excluded volume and sit near unperturbed dimensions, so a constructed cell’s C_n should be treated as an upper bound and re-measured after equilibration.
- L9. Linear chains only.** No branching, no polydispersity, no dissolved small molecules, no layered or oriented phases — all of which Materials Studio supports [10, 11].
- L10. Backbone rings are refused, not supported.** Back-mapping assumes the groups hanging off the backbone are *trees*. A ring *in* the backbone — poly(ethylene terephthalate), polycarbonate, PBAT — is a cycle spanning several skeletal atoms whose internal geometry is fixed by the ring, not by the RIS bond length and angle the growth used. The two are incompatible. **24 of the 123 usable library polymers** are of this kind. PAAF now detects them and raises a specific error; growth and coarse-grained export still work, since a bead is a skeletal atom either way, but no all-atom cell and therefore no force field is possible. Pendant rings (polystyrene) are unaffected.
- L11. The cell must be larger than twice the longest repeat unit.** Minimum image requires it. A repeat unit with a long side chain — an octyloxy group is 11.1 Å across — violates it in a small cell, and bonds measured across the boundary are then wrapped to the wrong periodic image. Measured for `[*]C(OCCCCCCCC)C[*]`: bonds up to 12.29 Å at 2 chains of DP = 6 (box 15.4 Å), and 1.093–1.532 Å for the same polymer at 4 chains of DP = 20 (box

29.0 Å). The structure was never wrong; the cell was too small to describe it. PAAF now compares the template extent against half the box and warns.

- L12. Some library SMILES layouts are not expanded reliably.** The n -mer expansion recognises the two common layouts, `[*]A[*]` and `[*]A([*])B`. Entries that put a marker inside a branch, use a bare `*`, or do not begin with a marker fall back to stripping the markers and repeating, which is not always correct, and log a warning when they do. Repeat-unit *masses* are unaffected because they need only one unit.

9 Relation to existing tools, and an honest novelty assessment

9.1 Comparison

	Materials Studio	PAAF
Growth method	Theodorou–Suter Meirovitch	+ same
Chain statistics	RIS	RIS, validated against literature C_∞
Resolution	full atomistic	united skeletal atoms, then back-mapped
Non-bonded during growth	production force field	generic soft repulsion
After construction	relaxes to potential-energy minimum	<i>nothing</i>
System types	polydisperse, branched, gases, layers, nematics	linear homopolymers and blends
Licence	proprietary	open, scriptable, seed-reproducible

Table 6: PAAF against BIOVIA Materials Studio Amorphous Cell [10, 11].

Among open-source tools, RadonPy [12] automates SMILES $\rightarrow$ amorphous cell $\rightarrow$ MD $\rightarrow$ properties, pysimm [13] provides polymer construction into LAMMPS, and SPACIER [14] builds on RadonPy for machine-learning-driven design. These construct cells predominantly by self-avoiding random walk with packing and compression, relying on extended equilibration to relax chain dimensions, rather than by RIS-weighted growth.

9.2 Novelty

Stated plainly: **the method implemented here is not novel.** It is a faithful reimplementa-tion of Theodorou–Suter (1985) and Meirovitch (1983) on Flory’s RIS foundation (1969). Nothing in the growth algorithm would survive a novelty assessment. The composition solver, the difficulty ordering and the back-mapping are competent engineering, not new science.

What is arguably underserved is the combination: *RIS-correct construction in an open-source tool*. Materials Studio has it and is closed; the widely used open tools largely do not. That suggests one testable claim, which would need to be demonstrated rather than asserted:

A cell built with RIS statistics begins nearer the correct chain dimensions and therefore reaches equilibrium in less MD time than a random-walk cell of identical composition.

Testing it requires building matched systems by both routes, running NPT equilibration, and tracking C_∞ and R_g against simulation time. If the crossover is substantial, that is a modest and publishable software result. If it is not, the tool’s value is convenience and transparency rather than physics — which is worth establishing before making any claim.

Two developments would constitute genuine contributions. The first, and stronger: **deriving RIS parameters automatically from SMILES** by torsion scanning a trimer fragment at MMFF, GFN2-xTB or DFT level and fitting E_σ , E_ω and the state angles. This would extend RIS-correct construction from four polymers to arbitrary chemistry and would remove limitation L4, which is the one that most restricts the method’s usefulness. The second: carrying the Rosenbluth weight through scanning so that dense cells and unbiased chain dimensions can be obtained simultaneously, removing limitation L6.

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